Prakash Chaudhary

Location: Seattle, WA **Phone:** +1-256-468-1243 **Email:** connectwithprakash@gmail.com **Website:** connectwithprakash.com **LinkedIn:** linkedin.com/in/connectwithprakash **GitHub:** github.com/connectwithprakash

Summary

Machine Learning Engineer with over 5 years of experience designing and deploying scalable end-to-end ML models, with expertise in building AI agents leveraging human-in-the-loop automation to drive measurable business impact.

Industry Experience

Fetch Rewards | Seattle, WA

Machine Learning Engineer - AI Labs | May 2024 - Present

- Developed and released internal alpha of **multimodal shopping agent** that answers shopping questions, plans personalized shopping lists, and maximizes offer value
- Launched **multi-agentic workflow** for product matching, improving match rates by **3.1-3.3 percentage points** (73.8%→76.9%) and unlocking **$165M in GMV**; achieved **15% full automation** with **LLM-as-a-Judge guardrails** at **97% accuracy**, automating **50+ hours of manual operations to 2 minutes**
- Developed **3-agent annotation workflow** using **LangGraph** to replace double-annotation with **human-LLM agreement system**, reducing 2nd annotator workload by **60%** while maintaining dataset quality through human tie-breaking on disagreement cases for product matching models
- **Fully automated RAG pipeline** to flag incorrect product matches at **95% recall**, protecting **~$800K in GMV annually**; replaced manual workflow, reducing daily effort from **8 hours to 15 minutes** and **saving 40 hours weekly**

Fusemachines Inc. | New York City, NY

Machine Learning Engineer | Feb 2020 - May 2022

- Developed **multimodal contrastive learning pipeline** combining image and time-series data for sales forecasting of new products without historical data
- Led efforts to build an ensemble model to classify fashion products into Kate Spade's four customer segments using computer vision and scipy optimize, achieving **62% accuracy**
- Developed an in-house recommendation system using **Factorization Machine** model, AWS EC2 and Lambda, achieving ranking metrics comparable to AWS's recommendation engine
- Implemented **Multi-Quantile RNN-based** sales forecasting model for Coach fashion, reducing stock outs by **30%**

Academic Experience

NASA-IMPACT | Huntsville, AL

Graduate Research Assistant | Aug 2022 - May 2024

- Developed a hierarchical document classification baseline model for Earth Science, achieving **46% exact match accuracy** across three hierarchical levels, enhancing document organization and retrieval

Fusemachines Inc. | New York City, NY

Lecturer & Teaching Assistant | Mar 2022 - Mar 2024

- Taught **50+ students** in Data Science, covering Python, Git, REST API, SQL, LLM, and MLOps as part of the AI Fellowship Latin America 2023
- Taught Computer Vision as a lecturer and assisted with Machine Learning and Deep Learning courses as a Teaching Assistant for the AI Fellowship Nepal in 2022

Education

University of Alabama in Huntsville | Huntsville, AL

Master of Science in Computer Science (Thesis) | Aug 2022 - May 2024 **GPA:** 4.0

- **Relevant Courses:** Deep Learning, Survey AI, Algorithm, Big Data Computing, Software Engineering
- **Awards:** Phi Kappa Phi Honor Society

MITx on edX | Online

MicroMasters in Statistics and Data Science | May 2020 - Aug 2021 **Grade:** B

- **Relevant Courses:** Probability, Statistics, Machine Learning, Data Analysis

Projects

Spectral Unmixing using Machine Learning | Mar 2023 - Oct 2023

A ML system for material characterization using spectral signal from Infrared (IR) spectroscopy.

- Developed an ML pipeline, including signal processing, feature engineering, and a multi-label neural network model, achieving over **95% F1 score** for predicting material composition from spectral signatures across diverse datasets

Foxhound Security Solution | Nov 2019 - Feb 2020

SaaS-based user's behavior anomaly detection and network event log analyzer built using distributed ML.

- Designed and developed an end-to-end continual machine learning system for anomaly detection and reasoning using Apache Spark for ETL and data analysis, an AutoEncoder architecture, and statistical analysis

Sound Source Localization | Dec 2018 - Aug 2019

An 8-microphone cubical mesh that records the azimuth and elevation of an incoming sound source.

- Constructed a localization system using C++ with the GCC-PHAT algorithm and microphone mesh, and created denoising AutoEncoder model using Keras to denoise speech and achieved **MAE of 1.5 degree** in 3D localization

Technical Skills

LLM & Agentic AI: LangGraph, LangChain, RAG, Multi-Agent Systems, Fine-Tuning (LoRA), RLHF, Prompt Engineering (Chain-of-Thought, ReAct), LLM-as-a-Judge Evaluation, Embeddings & Vector Search

ML/AI: Computer Vision, NLP, Time Series Forecasting, Recommendation Systems, Statistical Modeling

Frameworks & Tools: PyTorch, TensorFlow, Hugging Face Transformers, Distributed Training (DDP), Docker, Kubernetes

LLM Ops: Langfuse, DeepEval, OpenTelemetry, Model Serving (TorchServe, Ollama), Quantization

Infrastructure: AWS (SageMaker, EC2, Lambda, S3), Apache Spark, Kafka, Snowflake, Vector DBs (FAISS, Pinecone, Chroma), FastAPI

Programming: Python, Go, C++, SQL